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Multi-bit graphene-based bias-encoded metasurfaces for real-time terahertz wavefront shaping: From controllable orbital angular momentum generation toward arbitrary beam tailoring

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ABSTRACT

In this paper, for the first time, a new generation of multi-bit graphene-based bias-encoded metasurfaces (MGBMs) is proposed for real-time reflected wavefront manipulation at terahertz (THz) frequencies. The architecture of the designed MGBM is composed of several meta-atoms whose operational status can be independently switched between eight digital states of "000" – "111" in a real-time manner. By mere changing the distribution of chemical potentials through an external electronic source, the occupying meta-atoms can be dynamically combined in different gradient, spiral-like, and spiral-parabola-like coding sequences. Different from earlier works, the proposed MGBM can be re-programmed for accomplishing multiple outlandish missions from generation of vortex wavefronts carrying a controllable amount of orbital angular momentum (OAM) toward emitting multiple arbitrarily-oriented pencil beams at the same time. By exploiting the addition theorem and convolutional principle, multi-type functionalities, such as multiple pencil beams, multiple vortex beams with different topological charges, and multiple pencil–vortex beams are simultaneously realized. The full control of reflected wavefronts is corroborated with numerical simulations and theoretical predictions. By offering new attractive degrees of freedom to EM waves, various future directions are expected for the proposed versatile MGBM such as controllable displays, modern information systems, moving target detection, etc.

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1. Introduction

With the demand for new technologies, how to better control electromagnetic (EM) wavefronts has reached the peak of attention in recent years in assorted artificial platforms, such as surface plasmons [1], photonic crystals [2], transformation optics [3], metasurfaces [4], and optimized coil [5]. Based on a continuously enriched understanding of EM phenomena, the recent efforts have focused on developing some straightforward methods being able to fully manipulate the different properties of EM waves, including amplitude, phase, polarization, and linear/angular momentum. Orbital angular momentum (OAM) with a high potential application in encoding and multiplexing is an origin-independent angular momentum of an EM wave, which macroscopically unveils itself in the spatial properties of the beam and the shape of a helical or twisted wavefront [6]. The original concept of OAM of photons was pioneered by Allen et al. in 1992, where it is manifested as a helical phase front: $\exp(i\ell\varphi)$ in which ϕ is the azimuthal angle and ℓ is the topological charge, an unbounded integer indicating the repeating rate of 2π phase shifts azimuthally along the beam axis [7]. The optical vortices carry an OAM of $\ell\hbar$, making them helpful for a number of applications in communication systems [8]. OAM beams with different topological charges are mutually orthogonal in space, which allows them to be multiplexed together along the same beam axis and de-multiplexed with low crosstalk [9]. Recent achievements have demonstrated that OAM can be applied in the microwave to infrared frequency spectra, which among them the field of terahertz science and technology has been experiencing increasing levels of interest and so presents a myriad of opportunities for high-speed communications [10], fast imaging [11], spectroscopy, etc. There are a number of ways to realize OAM beams, such as those involving the use of spiral phase plates [12], spiral reflectors [13], computer-generated holograms [14], and antenna arrays [15]. Nevertheless, there are still many crucial problems of this research area waiting for solving so that only a few studies about generating THz vortex wavefronts have been

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reported so far. For example, these conventional OAM beam generators are bulky and thus restrict their employment in integrated THz devices [16]. More importantly, the ability to switch between different OAM modes enables fully accessing the benefits of OAM degree of freedom, where the future advancement of classical and quantum systems will require OAM routing flexibility and controllability [17]. Such restrictions along with the recent tremendous progress in metasurface land prompt us to seek a method necessary to generate tunable THz OAM beams in a real-time manner.

Metamaterials are often called as man-made materials consisting of periodically arranged subwavelength resonators, which provide unusual properties that lead to new capabilities in light-matter interactions [18]. Metasurfaces are two-dimensional (2D) version of more general volumetric metamaterials with subwavelength thickness, low losses, and simple fabrication whereby a myriad of diverse functionality are being developed ranging from perfect absorption to scattering wave manipulation [4]. By engineering the metal inclusions in different 2D layouts, they offer the possibility of a wide range of spectral property with extreme sensitivity to their geometric patterns. The idea of imparting a position-dependent custom phase to the reflected wavefront can be exploited to compensate for the phase difference between the incident and desired wave profile, e.g., linear, spiral, and parabola scattering phases, to generate anomalous reflections and refractions [19], OAM waves [20], spatial power dividers [21], and focusing beams [22]. Recently, T. Cui et al. has revolutionary introduced the concept of coding metasurfaces as a link between the physical metasurfaces, digital worlds, and information science whose EM responses can be specified by the coding patterns determining how the occupying coding particles are combined in a 2D array [23]. Compared to the traditional wave manipulation tools (e.g., transformation optics [24]), digital metasurfaces benefits from the advantage of being easier to analyze, design and fabricate while they exceptionally enable a broader range of wave-matter functionalities [25]. To date, most functional metasurfaces designed for a particular appeal are based on passive and static designs defined through geometrical parameters, such as particle shape, size, and array layout [26–28]. Indeed, their operational statuses remain unchanged once they are constructed even for frequency–dependent [29] or bifunctional metasurfaces [30,31]. As a result, one major bottleneck at present for advancing this field is the lack of efficient devices to allow dynamic tunability of the device functionality with time. For example, Zhang et al. proposed a method to design 3-bit coding metasurfaces to reach flexible controls of the circularly-polarized waves, forming spin-controlled multiple beams with different polarizations in free space, such as pencil beams and vortex beams carrying orbital angular momentum [32]. Nevertheless, implementing a dynamic control over the response of the designed metasurfaces was not addressed which dramatically hinders their practical application. Recently, several types of research have mainly focused on electronically tunable coding metasurfaces in the microwave frequency region, leaving plenty of rooms for researchers working at more challenging frequency domains [33–35]. To deploy their full potential, the THz coding metasurfaces should be designed in a reprogrammable manner in such a way that the operational status of each digital particle can be electrically controlled in a real-time manner. Despite the introduction of a few approaches for attaining controllable treatments at THz frequencies, the difficulty of reorganizing the geometries or modifying the comprising materials after fabrication makes re-programmable THz digital metasurfaces much more sophisticated than that of the microwave and millimeter wave regimes. Moreover, with the inherent nonlinearity associated with the comprising active materials along with their restriction to the laboratory samples (e.g., commercial PIN diodes), the dynamic THz wavefront shaping still remains as a critical challenge [36].

Graphene is a flat mono-atomic layer of Carbon atoms organized in a 2D honeycomb-like lattice [37]. The arbitrary control of surface electric conductivity of graphene through external biasing or chemical doping has enabled radically different photonic devices where it could be integrated with flexible substrates [38]. Moreover, because of having a strong interaction with EM fields, the possibility of exploiting the graphene-assisted devices has been conceptually assessed in diverse applications such as tunable/broadband absorption [39], photodetectors [40], antennas [41], polarization convertors [42], reflection/transmission active modulators [43–47], light-controlled conductive switching [48], ultra-sensitive plasmonic sensors [49], tunable metamaterial [50], shielding [51], and THz wave manipulation [52]. Carrasco et al. introduced a graphene-based reflective cell to be utilized in THz reflectarrays, or similarly to realize the well-known generalized law of reflection [53]. In 2015, a simple sectored circular reflective surface was proposed to generate vortex radio waves where the local reflection coefficients of the composing graphene reflective cells were controlled by changing their chemical potential and geometry [54]. A tunable anomalous refraction metasurface composed of periodically patterned graphene nanocrosses with the same geometry, but linearly varied orientations for circularly polarized waves was obtained in the infrared regime [55]. In 2017, Deng et al. reported a graphene-based plasmonic metasurface comprising of anisotropic elements, which can independently control different polarized EM waves with reasonably small losses in the THz regime [56]. By adjusting the geometric dimension of the patch elements, Zhang et al. presented a tuning scheme of the metasurface enabled with graphene gated layers to electrically control the EM beam reflection at THz frequency [57]. Nevertheless, the proposed device allows only the single-beam switch operating between normal and other particular reflection angles or the dual-beam switch operating between normal and other particular directions. In 2018, Zhang et al. proposed a novel metal–graphene hybrid metasurface operating at THz frequencies to steer the reflected waves into multiple beams in the far-field domain [58]. Recently, a multi-functional graphene metasurface based on Pancharatnam-Berry phase cells has been reported to control the spin angular momentum and orbital angular momentum of the reflected wavefronts in the THz regime [59]. However, despite claims made about the multi-functionality in all above-mentioned references, the beam steering, focusing and OAM generation has not been achieved in a real-time manner and the reported structures must be re-implemented, if our desired functionality changes. Indeed as they constituted from geometrically encoded particles, the afore-mentioned devices solely locked to specific operational status, so that the manipulation of EM waves is fixed once the design is completed. Benefiting from the tunable feature of graphene, we reported a real-time and broadband manipulation strategy of THz waves by introducing a multifunctional 1-bit digital metasurface [60]. More recently, we have introduced the concept of phase-encrypted metasurfaces whereby the information of messages can be illusorily recorded on multiple parity beams generated by a 1-bit graphene-based coding metasurface [61]. Although several functionalities were integrated within our designed metasurfaces; however, the limited number of possible modes (0/1) in 1-bit coding metasurfaces inevitably restricts their effectiveness in generating asymmetric multi-beam radiations, and providing high capacity for carrying information. As a result, our 1-bit coding metasurfaces cannot address some crucial THz challenges such as anomalous reflection governed by the generalized Snell's law and OAM beam generation with tunable properties. Apparently, 2- and

3-bit coding metasurfaces have much more freedom in controlling EM waves than their 1-bit counterparts at the expense of a more difficult design. Although multi-functional fixed-function graphene-based structures have been demonstrated [53–64], we are not aware of any reports of real-time THz wavefront manipulation that reside on the reconfigurable multi-bit coding metasurfaces.

Beyond the scope of fixed-function coding metasurfaces, a versatile multi-bit graphene-based bias-encoded metasurface (MGBM) is designed in this paper, comprising digitally programmable meta-particles whose operational statuses can be instantaneously switched among eight states of "000", "001", "010", "011", "100", "101", "110", and "111". Among all possible configurations where metasurfaces can be applied, the focus of our study is concentrated on the analysis of reflective coding metasurfaces. Each constitutive meta-atom of the MGBM integrates two square-shaped graphene surfaces fed by distinct biasing voltages, whereby it can dynamically possess arbitrary phase responses of 0, $\pi/4$, $\pi/2$, $3\pi/4$, π, $5\pi/4$, $3\pi/2$, and $7\pi/4$ without any change in its geometry. In order to unleash the full potential of our design, we have designed different coding samples to do different interchangeable missions from the generation of vortex wavefronts with desired parameters toward arbitrarily-directed pencil beam radiation without any changes in the structure. Besides, it is demonstrated that a complete and accurate control on both the azimuth and elevation directions of the anomalously reflected beams can be dynamically achieved via establishing a proper phase-gradient sequence. By performing the convolutional and superposition theorems on the far-field pattern, multi-type functionalities, such as multiple pencil beams, multiple vortex beams with different topological charges, and multiple pencil–vortex beams are simultaneously realized. By this merit, we have more freedom in manipulating the EM energies by a new class of multi-bit re-programmable THz bias-encoded metasurfaces. The presented concepts and physical phenomena have been corroborated through a number of numerical simulations and theoretical predictions. Our work addresses several critical issues associated with the current stage of THz coding metasurfaces, featuring highly integration capability, being rewritable, conformal and high efficiency. To the best of our insight, this is the first re-programmable multi-bit coding metasurface that can provide switchable phase modulations at THz frequencies, carrying out various missions assigned to it in a real-time manner.

2. Materials and principles

2.1. Graphene's surface conductivity

The graphene material can be utilized to manipulate the THz waves because it is found to be extremely sensitive to external stimulus due to its boundary conditions. The AC conductivity of graphene can be tuned via external electrostatic bias or external magnetostatic bias by Hall Effect. Owing to its mono-atomic specification, graphene can be locally represented by a complex surface conductivity tensor [65].

$$\sigma(\omega, \mu_C(\mathbf{E}_0), \Gamma, T, \mathbf{B}_0) = \hat{\mathbf{x}}\hat{\mathbf{x}}\sigma_{xx} + \hat{\mathbf{x}}\hat{\mathbf{y}}\sigma_{xy} + \hat{\mathbf{y}}\hat{\mathbf{x}}\sigma_{yx} + \hat{\mathbf{y}}\hat{\mathbf{y}}\sigma_{yy} \quad (1)$$

In the absence of magnetostatic bias, the off-diagonal components of the surface conductivity tensor vanish, and isotropic behaviors are noticed by the graphene monolayer. The graphene monolayer can be electrically modeled as an infinitesimally thin conductive layer characterized by a complex-valued surface conductivity [65].

$$\sigma_S(\omega, \mu_c, \Gamma, T) = \sigma_S^{intra}(\omega, \mu_c, \Gamma, T) + \sigma_S^{inter}(\omega, \mu_c, \Gamma, T) \quad (2)$$

where, σ_S^{intra} and σ_S^{inter} represent intraband and interband parts of surface conductivity, respectively. According to the Kubo formula, each of the conductivity portions can be calculated as follows [65]:

$$\sigma_S^{intra}(\omega, \mu_c, \Gamma, T) = -j\frac{e^2 k_B T}{\pi\hbar^2(\omega - j2\Gamma)}\left[\frac{\mu_c}{k_B T} + 2\ln\left(e^{-\mu_c/k_B T} + 1\right)\right] \quad (3)$$

$$\sigma_S^{inter}(\omega, \mu_c, \Gamma, T) = -j\frac{e^2}{4\pi\hbar}\left(\frac{2|\mu_c| - (\omega - j2\Gamma)\hbar}{2|\mu_c| + (\omega - j2\Gamma)\hbar}\right) \quad (4)$$

in which, $\hbar = h/2\pi$ is the reduced Planck's constant, K_b is the Boltzmann's constant, e denotes the electron charge, τ is the transport relaxation time, $\Gamma = 1/2\tau$ is the scattering rate, T is the environmental temperature, ω is the operating angular frequency, and μ_c is the chemical potential which can be dynamically controlled by applying an external electric field. At room temperature and low terahertz frequency, ($\hbar\omega \leq 2\mu_C$), the transitions of interband were forbidden by the Pauli Exclusion Principle [66]. In the following simulations, the two parameters of temperature and transport relaxation time are considered constant and equal to $T = 300$ K and $\tau = 0.3$ ps respectively, which are in agreement with measured data from the chemical vapor deposited (CVD) graphene [67]. Moreover, the dielectric constant of graphene layer is considered as $\varepsilon_s = 1 + j\sigma_s/\omega\varepsilon_0\Delta$ wherein Δ is a very small value indicating the thickness of the graphene layer. Importantly, the surface conductivity of the graphene surface can be efficiently manipulated via a perpendicular bias electric field, enabling the possibility of dynamic switching between all different missions assigned to the designed MGBM in this paper. The detailed information can be found in Supplementary Information A.

2.2. Coding element design and its circuital representation

Although employing the coding metasurfaces has facilitated the wavefront transformations, current endeavors to implement them still lead to incorporating multiple elements of different geometries or orientations which makes the design procedure difficult and static. Therefore, the multi-bit metasurface designs encounter with time-consuming optimization steps where the associated computational burden increases exponentially with the number of coding particles. Most of the current design schemes rely on full-wave numerical simulations to associate a given subwavelength element with its equivalent digital status, yielding a complex lookup table that is established for coding metasurface realization. Conversely, the primary objective of this work is to design a versatile MGBM meta-device in which the operational status of each occupying lattice can be individually set via a multi-voltage DC electronic source, providing flexible and real-time control over reflected wavefront at THz frequencies (see Fig. 1). One of the most significant drawbacks of single layer meta-particles is their restricted interactions with the incident EM waves, which impose severe limitations on their EM responses [68]. By contributing to a broader phase spanning the range between [0,2π), the Fabry-Perot resonance induced by few-layered inclusions makes them a good alternative to tackle this challenge. Moreover, the slow wave propagation supported by graphene dramatically diminishes the meta-atom dimensions, leading to a smaller inter-element spacing and a more compact design. Accordingly, the constitutive meta-atoms of the proposed MGBM integrate two square-shaped graphene surfaces fed by distinct DC voltages, whereby each of them

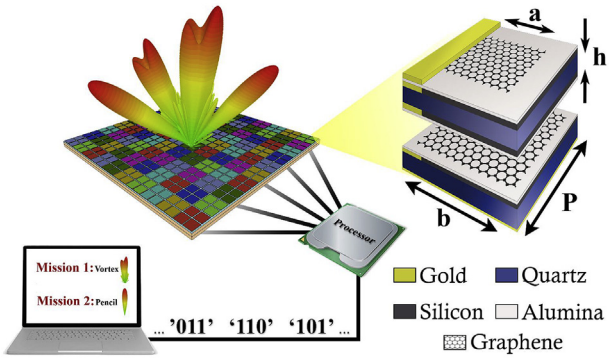

Fig. 1. A sketch of the re-programmable MGBM composed of an array of meta-atoms whose operational status can be dynamically altered by a field programmable gate array (FPGA). A computer generating different pre-determined coding sequences can digitally drive the MGBM layout. Armed with different reflection phase maps, the MGBM can successfully accomplish multiple missions at the far-field domain. The perspective of the meta-atoms can be found in the upper right corner. (A colour version of this figure can be viewed online.)

can dynamically possess arbitrary phase responses of 0, $\pi/4$, $\pi/2$, $3\pi/4$, π, $5\pi/4$, $3\pi/2$, and $7\pi/4$ to mimic the '000', '001', '010', '011', '100', '101', '110' and '111' elements, respectively. The employed graphene-based meta-atoms in this paper have the same size but different chemical potentials. Therefore, their identical geometries allow us to call them as *bias-encoded* meta-atoms, not *geometrically-encoded* ones. As the inset of Fig. 1 shows, the employed meta-atom is a stack of a metal layer, Quartz dielectrics, and floating-gates based on two square-shaped graphene metasurfaces. The graphene patterns are located on top of two identical Quartz substrates ($\varepsilon_r = 3.75$, $\tan\delta = 0.0184$) with the thicknesses of $h = 12.2\ \mu m$. The upper and lower graphene metasurfaces have the length of $a = 11.1\ \mu m$ and $b = 15.7\ \mu m$, respectively. The EM mutual coupling between the top and bottom graphene layers have been comprehensively studied in Supplementary Information B. Each meta-particle has the periodicity of $P = 15.9\ \mu m$ and is terminated by a gold plane to prevent the transmission of energy into the rear of GBM. Due to their geometrical symmetry, the sub-wavelength graphene-based building meta-atoms offer a polarization-independent performance upon illumination by the incident waves of different polarizations. To enable the tunability of the charge density corresponding to each patterned graphene layer, the floating-gate structures are utilized here. As depicted in Fig. 1, the floating-gate panels are composed of polycrystalline silicon (poly-Si), Al_2O_3, and square-shaped monolayer graphene. In each of them, when a positive voltage is applied on the top poly-Si layer, the electrons in the bottom ploy-Si layer can tunnel through the Al_2O_3 and be captured by the graphene metasurface, yielding higher charge density and chemical potential [69]. On the contrary, when the reverse voltage is applied on the top poly-Si layer, the electrons in graphene can tunnel through the Al_2O_3 and be captured by the bottom poly-Si layer, leading to the decrease in its charge density and chemical potential. Moreover, because of being isolated from the poly-Si layers, the chemical potentials of graphene metasurfaces can remain constant for a long period after the external biasing voltages are disconnected. Further practical considerations regarding the bias method are given in the Supplementary Information A. The thickness of auxiliary layers is much smaller than the operating wavelength so that their impact on the phase and amplitude of reflection spectrum can be safely ignored. The process of sample fabrication may follow the steps below, and so one can envision a feasible fabrication within the realm of the current fabrication technologies. Firstly, using spin-coating solution, the Quartz layer can be coated on a silicon wafer. The graphene samples can be fabricated using CVD graphene grown on Cu foil and transferred onto a spacer of 65 nm-thick insulating (Al_2O_3) layer that was uniformly deposited by atomic layer deposition on an ultra-high resistivity p-type poly-Si wafer. After being transferred to Al_2O_3/poly-Si wafers, the excess graphene was removed in the sheets using 100 keV electron beam lithography in PMMA followed by an oxygen plasma etch, leaving only the square-shaped regions. The transfer and dielectric processes are then repeated for the second layer. Indeed, another Quartz substrate used as a supporting layer during the transfer process is kept on top of the graphene to act as a dielectric between two graphene screens. The second patterned graphene is subsequently transferred onto the top of the insulating (Al_2O_3) layer. Finally, the bottom silicon wafer can be peeled off using KHF solution with metal deposited on the back of Quartz layer as a ground layer, thus obtaining the final structure shown in Fig. 1. To further reduce the access resistance of the device, the metal electrodes are patterned by optical lithography followed by a deposition of 5 nm of chromium on top of the graphene layers, 50 nm of gold and a lift-off process. Exploiting a controllable capacitor consisting of a thin layer of Al_2O_3 as the dielectric spacer and poly-Si layer as the back-gate electrode cause the chemical potential of graphene layers to be dynamically tuned in the range of 0–1.3 eV. The geometrical configurations of different stacking sequences gets more rich physics due to the interlayer couplings. Therefore, the potentially limited phase modulation of a single graphene layer can be circumvented by the stacked structure where each layer can efficiently contribute to a broader phase tuning range [68,70]. The reflection characteristics have been assessed for each meta-particle via commercial full-wave software, CST Microwave Studio, including the coupling effect between neighboring elements. Periodic boundary conditions are applied in the *x* and *y* directions while Floquet ports are assigned to the *z*-direction. The meta-atoms occupying the re-programmable 1-bit, 2-bit, and 3-bit graphene-based bias-encoded metasurfaces (GBMs) are parametrically tailored to provide a certain reflection phase upon illumination by a normal plane wave. The resultant reflection spectra are comprehensively illustrated in Fig. 2 for different sets of chemical potentials. As the inset of this figure shows, two anti-phase graphene-based units having the chemical potentials of 0.65eV/0.1eV (the first value represents the chemical potential for the upper graphene layer, and the latter refers to that of the bottom layer) and 0.05eV/0.7eV can be established as the 1-bit coding particles "0" and "1", respectively.

Those graphene-based elements with the chemical potentials of 0.7eV/0eV, 0eV/0.2eV, 0eV/0.75eV, and 0.3eV/1eV, which provide a phase step of 90°, can mimic "00", "01", "10", and "11" states, respectively. Finally, the meta-particles with the chemical potentials of 0.55eV/0eV, 0.75eV/0.25eV, 1.2eV/0.5eV, 0.15eV/0.05eV, 0eV/0.4eV, 0eV/0.7eV, 0.2eV/1eV, and 0.35eV/1.3eV, with a reflection phase response of 45° ahead to the previous state, can be exploited to realize the 3-bit coding particles "000", "001", "010", "011", "100", "101", "110", and "111", respectively. The summary of the above discussion can be found in Table 1. It is worth mentioning that due to the weaker interaction between the incident waves and electrons, the plasmonic response of noble metals becomes less pronounced as operating in THz bands [71]. The key point is that all our designed coding elements depict a high reflectivity over the operating bandwidth as they function far from their resonance frequency. Amazingly, the reflection phase responses have a nearly linear trend, which is very crucial for designing multi-bit bias-encoded meta-atoms with identical geometries. It can be seen from Fig. 2 a,b that the perfect binary state occurs at 2 THz, where the states "0" and "1" possess identical reflective amplitudes and opposite reflective phases over a broad frequency range between

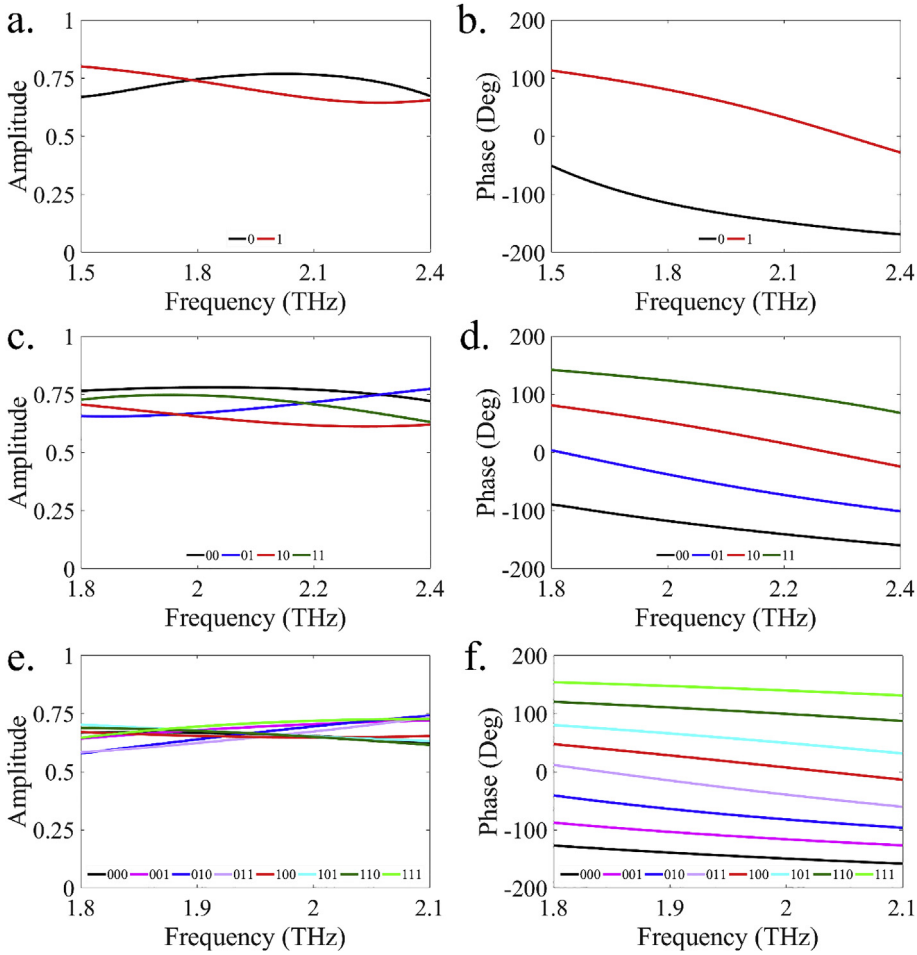

Fig. 2. The simulated reflection-phase (b), (d), (f) and reflection-amplitude (a), (c), (e) spectra of the employed digital meta-atoms. (a), (b) "0" and "1" digital meta-atoms. (c), (d) "00", "01", "10", and "11" digital meta-atoms. (e), (f) "000", "001", "010","011", "100", "101", "110", and "111" digital meta-atoms. (A colour version of this figure can be viewed online.)

Table 1
The required chemical potentials for realizing each coding particle occupying the versatile MGBM.

ϕ_{ref}	0	π/4	π/2	3π/4	π	5π/4	3π/2	7π/4		
1-bit MGBM	0.65				0.05				1st layer	Chemical Potential (μ_c)
	0.1				0.7				2nd layer	
	0				1				Code	
2-bit MGBM	0.7		0		0		0.3		1st layer	
	0		0.2		0.75		1		2nd layer	
	00		01		10		11		Code	
3-bit MGBM	0.55	0.75	1.2	0.15	0	0	0.2	0.35	1st layer	
	0	0.25	0.5	0.05	0.4	0.7	1	1.3	2nd layer	
	000	001	010	011	100	101	110	111	Code	

1.5 and 2.4 THz. As shown in Fig. 2, by increasing the number of metasurface bits, the frequency bandwidth of the coding elements is approaching to 1.8 THz to 2.1 THz; however, a linear phase change is still available in the whole frequency band of interest. Here, the full phase range of 360° cannot be available since the intrinsic losses of the graphene patch make the structure operate as a damped oscillator [72]. Nevertheless, a maximum phase range of 330° is acquired here that is wide enough for practical applications requiring an acceptable efficiency. Therefore, the cell configurations corresponding to Fig. 2 a,b, Fig. 2 c,d, and Fig. 2 e,f can be considered as 1-bit, 2-bit, and 3-bit bias-encoded building meta-atoms, respectively. Using a proper electrostatic bias circuit, the digital status of each meta-particle can be dynamically toggled between two, four or eight possible modes, providing an unprecedented degree of freedom in real-time wavefront manipulation of THz waves.

It should be remembered that the different bias voltages solely cause the change in the reflection phase responses of neighboring coding elements. Evidently, the operating bandwidth can be further broadened via modifying the geometry and shape of the graphene-based particles so that the desired reflection phases can be acquired in a broader frequency range. Prosperously, the

collective response of a given coding pattern is not affected by the change of individual coding elements. Indeed, any arbitrary digital layout has unique far- and near-field patterns solely determined by its coding sequence and vice versa, which is of a great benefit to the re-programmable MGBM. They can be fruitfully represented by a physical equivalent circuit in which each patterned graphene between two stratified media (air-quartz or quartz-quartz) is represented as an R-L-C branch (see Supplementary Figure S4). The supporting substrates can be modeled as distinct transmission lines with the intrinsic impedance, and propagation constant of $\eta_c = \eta_0/\sqrt{\varepsilon_r}$ and $\gamma = \omega\sqrt{\varepsilon_r}/c$, respectively wherein c and η_0 refer to the speed of light and the free-space characteristic impedance. The metasurface particles backed with a ground plane can be theoretically modeled with a short-terminated circuit. The values of the circuit components have been extracted from the reflection and transmission scattering parameters of the graphene metasurfaces at the boundary between air and substrate [73]. For the sake of brevity, we only consider the 1-bit coding elements whose phase and amplitude response can be calculated by the proposed circuit model. The extracted R, L, and C values are listed in Supplementary Table S1 for various representative cases. Supplementary Figure S5 compares the reflection spectra obtained by the circuit model with those of full-wave numerical simulations where an excellent agreement is noticed in both phase and amplitude responses. The trends are physically intuitive and can be elaborately justified by the evolution of the real and imaginary parts of graphene conductivity as well as the electrostatic capacitive coupling between the neighboring elements [53]. Therefore, establishing the circuit model promotes the following design process and the corresponding results are in good agreement with the expectancy.

3. Results and discussion

In the following, we answer this question that *"Can we have a dynamically re-programmable multi-bit coding metasurface by abruptly varying the graphene local conductivity, which allows challenging the usual reflection response at a transversely homogeneous interface?"*. Under the illumination of THz incidences, all digital meta-atoms are driven by a certain coding sequence, resulting in a special type of far-field pattern whose information can be dynamically modulated via external biasing voltages. In this case, each meta-particle can be treated as a variable contributing to the total radiation. Once the designed MGBM has been prepared, merely changing the biasing voltages connected to each meta-particle can re-program the metasurface with different "coding patterns" of

$$M_{1-bit} = \begin{pmatrix} 0 & 1 & \cdots & 1 \\ & \ddots & & 0 \\ 1 & & 0 & \vdots \\ 0 & \cdots & & 1 \end{pmatrix}, M_{2-bit} = \begin{pmatrix} 00 & & \cdots & \\ & \ddots & & 11 \\ 11 & & 10 & \vdots \\ & 01 & \cdots & 00 \end{pmatrix} \text{ and } M_{3-bit} = \begin{pmatrix} 000 & & \cdots & \\ & \ddots & & 010 \\ 111 & & 110 & \vdots \\ 001 & \cdots & & 100 \end{pmatrix} \quad (5)$$

When all possible coding sequences are stored in an FPGA, the spatial arrangement of two, four or eight bias-encoded meta-particles on the surface plays an enormous role in determining EM functionality of the multi-mission MGBM.

3.1. Re-programmable 1-bit GBM: symmetrical beam deflection, scattering diffusion

To begin with, a simple type of coding metasurfaces is considered, whose coding pattern consists of only two interchangeable coding elements of "0" and "1" with the reflection phase responses of $\varphi_n = n\pi$ (n = 0,1). To minimize the EM coupling between the elements in adjacent lattices, the MGBM structure consists of $M \times M$ lattices occupied with 10×10 identical meta-particles throughout the paper, yielding the lattice constant of $D_x = D_y = 159$ μm. Because of a direct connection between different coding patterns of the metasurface and their radiation patterns, under normal incidence, the scattering pattern of the MGBM can be expressed in the far-field domain by Ref. [74].

$$E_{scat}(\theta,\phi) = \sum_{m=0}^{M-1}\sum_{n=0}^{M-1} f_{m,n}(\theta,\phi).A_{m,n}\exp j(mu+nv) \quad (6)$$

$$f_{m,n}(\theta,\phi) = \left(\hat{\mathbf{y}}\cos\theta - \hat{\mathbf{z}}\sin\theta\sin\phi\right) \times \mathrm{sinc}\left(\frac{u}{2}\right)\mathrm{sinc}\left(\frac{v}{2}\right) \quad (7)$$

where $u = k_0D_x\sin\theta\cos\varphi$, $v = k_0D_y\sin\theta\sin\varphi$, $A_{m,n} = a_{m,n}\exp(j\varphi_{m,n})$ is the complex reflection coefficient, $f_{m,n}(\theta,\varphi)$ is the pattern function of each lattice, $\mathrm{sinc}(x) = \sin(x)/x$, k_0 is the free-space wavenumber, and θ and φ are elevation and azimuth angles, respectively. Since the isotropic radiation of each lattice arises from its sub-wavelength dimensions, we can safely ignore $f_{m,n}(\theta,\varphi)$ during the calculation of the far-field pattern. This formulation of the well-known antenna array theory is useful for the prediction of scattering patterns caused by different coding sequences. Nevertheless, by increasing the number of lattices within the whole metasurface, the computational burden arising from the low convergence rate turns Equation (6) into an annoying post-processing step, since there are approximately 2^{bM^2} possibilities for the coding pattern where b denotes the number of coding bits. To circumvent this limitation, Equation (6) can be re-expressed as a 2D inverse fast Fourier transform (2D-IFFT), to significantly accelerate the calculation [75], which is very promising for further designing steps. The related details can be found in Supplementary Information D. To demonstrate the capabilities of the designed 1-bit GBM meta-device, three different samples driven by distinct coding sequences of 01010 …/01010 …, 1010 …/0101 …, and a pseudo-random computer-generated coding pattern are considered here. The pseudorandom code is simply obtained in the same manner explained in Ref. [60]. In subsequent, each arrangement is normally illuminated by an x-polarized normally incident wave, and the corresponding 3D scattering patterns are depicted in Fig. 3, at $f = 2$ THz. Owing to the inherent geometric symmetry in the designed meta-particles, the EM responses of the MGBM structure is almost independent of the polarization states of the incident THz waves. We first demonstrate the beam-splitting performance of our re-programmable MGBM by endowing the surface with 01010 …/01010 coding sequence programmed by dual-voltage external source imparted on the graphene-based meta-particles. From Fig. 3a, one can conclude that such a periodic striped coding sequence leads to two symmetrically-oriented scattered beams pointing to same elevation directions governed by the generalized Snell's law. The GBM meta-device of Fig. 3b in which the "0" and "1" coding elements are engineered in a chessboard configuration will squeeze the incident wavefront into four pre-determined directions of the same elevation angles. Finally, to implement a reflective GBM with diffusion-like backward scattering, we uniformly extend the "0" and "1" coding elements in a finite surface of 10×10 lattices according to a computer-generated pseudorandom

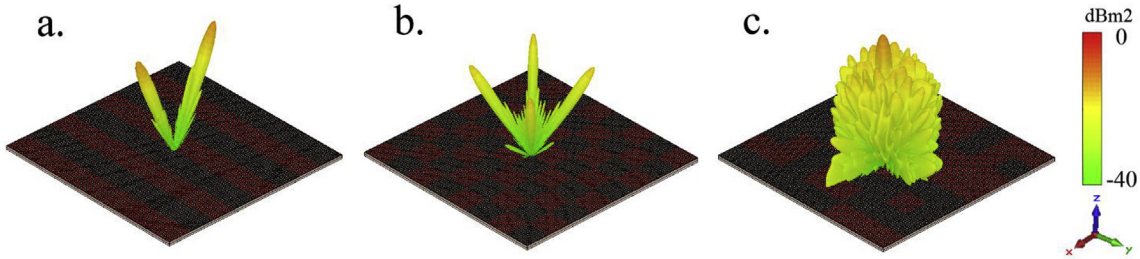

Fig. 3. Full-wave simulation results of backward 3D scattering patterns at $f = 2$ THz for the proposed 1-bit GBM meta-device driven by different coding sequences under the y-polarized normal incidences. (a) 010101 … /010101 …, (b) 010101 … /101010 …, and (c) a computer-generated pseudorandom coding sequence. (A colour version of this figure can be viewed online.)

coding sequence. Fig. 3c shows that randomized GBM can significantly suppress the scattered wavefront in all directions in low levels. Indeed, the occupying digital lattices are remarkably disordered in such a manner to redirect the impinging THz waves into many randomly directed beams, in each of which, the energy is small based on the energy conservation principle. Such an interesting phenomenon called *diffuse scattering* clearly discards the dominating specular reflection for a bare metallic reference plate of the same size. Eventually, it is shown that the 1-bit GBM structure can carry out different missions such as symmetrical beam deflection and scattering diffusion. However, the limited number of possible modes (0/1) in 1-bit coding metasurfaces inevitably restricts their effectiveness in generating asymmetric multi-beam radiations and providing high capacity for carrying information. Consequently, the metasurface should be driven by a multi-bit coding pattern to be capable of producing more complex functionalities such as single-beam asymmetric reflection and OAM wavefront of ring-shaped far-field intensity.

3.2. Re-programmable versatile MGBM

The concept of coding metasurfaces can be extended from 1-bit configurations to multi-bit ones, which provide greater flexibility in drawing into various space-varied phase distributions at the expense of more sophisticated design and complex technology for digital control. In this paper, any desired reflection phase map is digitally available by merely changing the multi-level biasing voltages feeding each meta-particle of the MGBM.

3.2.1. Phase-gradient coding sequence: single-beam modulation

As an interesting example, phase gradient coding sequences create artificial surfaces that can provide pre-defined in-plane wave-vectors to arbitrary tailor the direction of the reflected wavefront. The incident plane wave is either efficiently trapped due to surface wave coupling or is deflected to other directions because of the anomalous reflection. For the sake of briefness, the latter case is only studied in this paper. Based on the lateral phase gradient, a suitably engineered transverse phase discontinuity profile imparted over the MGBM can divert an impinging wave into a new predetermined route. Suppose EM waves are illuminating the MGBM structure with a predefined propagation vector $\vec{k}_i$. In the framework of 2-bit coding metasurfaces, the phase responses φ_n are simply defined as $\varphi_n = n\pi/2$, (n=0, 1, 2, 3). The phase gradient coding sequences (0 1 2 3 0 1 2 3 …) in the two horizontal and vertical directions, $\nabla\varphi_x$ and $\nabla\varphi_y$, are designed to provide the additional wave-vectors along the two in-plane directions (see Fig. 4). The 2-bit MGBM is composed of several superlattices with the periodicity of $L_x = N_x D_x$, and $L_y = N_y D_y$ each of which is occupied with $N_x = N_y = 2^2$ lattices along x and y directions, respectively. The MGBM can be driven by either the coding sequence of

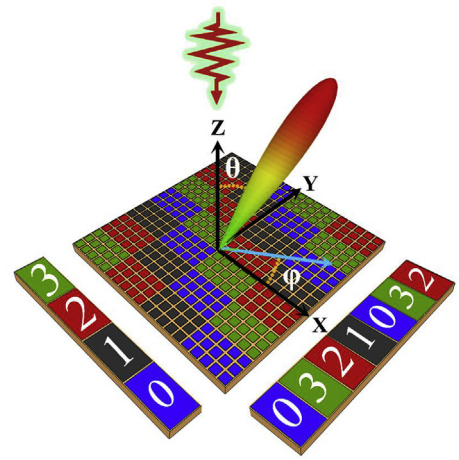

Fig. 4. The illustration of real-time single-beam deflection into an arbitrary predetermined direction upon illumination by a normal plane wave. The MGBM is suitably programmed with different switchable gradient coding sequences (0 1 2 3 …) varying along both vertical and horizontal directions. (A colour version of this figure can be viewed online.)

11→10→01→00 (a gradually descending reflection phase distribution with the step of $\Delta\varphi_{x/y} = -\pi/2$) or 00→01→10→11 (a gradually ascending reflection phase distribution with the step of $\Delta\varphi_{x/y} = \pi/2$). We define values $\nabla\varphi_x = \Delta\varphi_x/D_x$ and $\nabla\varphi_y = \Delta\varphi_y/D_y$ as the slope of lattice phase variations along the x and y directions, respectively. In this case, the in-plane wave-vector component of reflected waves is not equal to that of the incident waves; instead, we have $k_r^{\parallel} = (k_{ix} + \nabla\varphi_x)\hat{u}_x + (k_{iy} + \nabla\varphi_y)\hat{u}_y$ wherein $k_{ix} = k_0 \sin\theta_{inc}\cos\varphi_{inc}$ and $k_{iy} = k_0 \sin\theta_{inc}\sin\varphi_{inc}$. For $k_r^{\parallel} > k_i$, the interesting anomalous reflection phenomenon appears in which the direction of the scattered wavefront can be arbitrarily modulated based on the generalized Snell's law [19].

$$\theta_{ref} = \arcsin\left(\frac{\sqrt{k_{rx}^2 + k_{ry}^2}}{k_0}\right) \tag{8a}$$

$$\phi_{ref} = \arctan\left(\frac{k_{ry}}{k_{rx}}\right) \tag{8b}$$

Here, θ_{ref} and φ_{ref} refer to the azimuth and elevation angles of the reflected wavefront, respectively. Obviously, by proper designing the coding sequences, the reflected beam can be directed toward a predetermined angle in each of four quadrants of the upper half-space. The beam deflection capability of the phase gradient MGBM is demonstrated and exemplified in Supplementary Information E. Numerical simulations of Supplementary Table S2 show excellent

agreements with our theoretical calculations, highlighting the first important application of our design: real-time asymmetric beam deflection towards arbitrary angles, which cannot be achieved by conventional fixed-function coding metasurfaces [76–79]. Theoretically speaking, the slight deviation of the results for high deflection angles can be attributed to the local impedance mismatch between the incident and the desired reflected wavefronts [80]. The quantitative results describing the beam deflection efficiency of the proposed phase gradient MGBM are given in Supplementary Information E. More beam–deflection angles could be expected through the ingenious design of the voltage configuration imparted on the patterned graphene surfaces. Besides, it should be noted that regarding Equations (8a) and (8b), one could envision a beam steering functionality at the same frequency by changing the size of lattices of the designed MGBM meta-device by varying the distribution of the digital codes. Eventually, by applying a multi-voltage external source to the gated structure controlled by an FPGA platform, the instantaneous access to a single beam pointing at any desired angles is simply available, which may be of great interest in many applications such as meta-antennas, computational imaging, and tracking systems. In a one-dimensional scenario ($k_y = 0$), Equation (8a) turns into $\theta_{ref} = \sin^{-1}(\lambda/L_x)$. As long as $\lambda \leq L_x$, the maximum deflection angle ($\theta_{ref,\max}$) is resulted by the minimum length of superlattices, $L_{\min} = M_{\min} \times P$. With assuming $P = 15.9\ \mu m$ and $f = 2$ THz, we have $\theta_{ref,\max} = \arcsin(9.43/M_{\min})$ from which the minimum number of meta-atoms satisfying $\sin\theta_{ref} \leq 1$ is $M_{\min} = 10$, resulting in the maximum deflection angle of 70.6°. Besides, the step size achievable (in the control of deflection) using the proposed phase gradient MGBM is presented in Supplementary Information E.

3.2.2. Spiral–Parabola coding sequence: focused OAM beam generation

To further demonstrate the versatility of the designed MGBM, it is encoded by a spiral coding sequence to produce vortex beams upon illumination by a normally incident linearly polarized wave. By suitably arranging the subwavelength meta-particles with different operational statuses, the impinging EM waves on the metasurface can be translated into the one with the OAM mode, and the total phase range of the metasurface determines the topological charge (ℓ) of OAMs. Given that converting the incident plane wave into the vortex wave cannot be efficiently achieved by 1-bit GBM, so a 3-bit reflective MGBM is employed here. The beam modulating process is carried out by N sectors all of which are required to yield the same reflectivity with phase variation incremented by a multiple integral ℓ of 2π upon a full rotation. The resultant spiral phase distribution generating a vortex beam with OAM mode ℓ can be written as below.

$$\varphi(x,y) = \ell \times \theta = \ell \times arctan(y/x) \tag{9}$$

Here, θ is the azimuth angle indicating the position of each meta-particle while x and y are the distances between the centers of each meta-particle and the MGBM structure. The twisting of the phase into a corkscrew shape yields a doughnut-like ring intensity profile. After calculating the required 2D reflection phase map, each value is quantized to a 3-bit digital state implemented by one of the eight designed meta-particles in Table 1. To illustrate the reprogrammable property of the MGBM meta-device, four OAM-generating samples with different OAM modes ($\ell = -2, -1, 1, 2$) are presented and illuminated by a linearly polarized normal incidence to validate the concept. In each case, the 2D reflection phase map and the required chemical potentials for realizing the corresponding digital states are given in Supplementary Figure S15. Due to the limitation of the computer resources, the instant electric field intensities are recorded in a near-field sampling plane only 200 μm away from the center of finite GBMP with an area of 1590 μm × 1590 μm (see Fig. 5a-d). Generally, rendering a wideband OAM launcher seems impossible as the phase discontinuity induced by each section of the metasurface cannot remain the same if the operating frequency is changed. Nevertheless, by keeping the decremental phase shift equal to $2\pi/N$, a wide operating bandwidth for generating OAM vortex beams is obtained here from 1.8 to 2.1 THz. To demonstrate the working bandwidth of the OAM generation, the results are captured at different frequencies of $f = 1.8$ THz, $f = 1.9$ THz, $f = 2$ THz, and $f = 2.1$ THz, respectively. Corresponding to the forecasts from the analysis, with reference to the figures, the mode number of vortex beams is conspicuous. The striking features of the designed wave-shaping MGBM may be better appreciated by investigating the hollow radiation patterns plotted in Fig. 5e-h, corresponding to the designs in Supplementary Figures S15.a-d. It is shown that as the topological charge increases from $|\ell| = 0$ to $|\ell| = 2$, the reflected electric field intensity at the center area decreases so that the amplitude nulls in OAM modes $\ell = \pm 1$ and $\ell = \pm 2$, can be clearly distinguished. Both the 3D far-field scattering pattern with a hollow in the center and the spiral-type phase accumulation demonstrate that the MGBM can efficiently produce vortex beams carrying OAM ($\ell = -2, -1, 1, 2$). Amazingly, the designed meta-device has the ability to switch between different OAM modes in a real-time manner. This feature can enable dramatic networking advances where it might prove useful for information encoding, communication capacity enhancement without broadening the bandwidth, and OAM-based multiuser systems for which the OAM value of the beam determines the routing.

The versatility of the proposed MGBM also provides the ability to focus the incident EM wave into a pre-determined point, which significantly increases the power intensity of the generated vortex wave [6]. To obtain a focused wavefront carrying OAM, a parabola phase distribution along the radial direction should be involved in Equation (9), making the scattered fields constructively interfere at the pre-determined focal point [6].

$$\varphi(x,y) = \ell \times arctan(y/x) + \frac{2\pi}{\lambda}\left(\sqrt{x^2 + y^2 + z_{focal}^2} - z_{focal}\right) \tag{10}$$

where λ denotes the operating wavelength, and z_{focal} refers to the focal length along the z-direction. One can deduce that the phase distribution contains two main function components one of which is the spiral phase distribution (the first term in the right-hand side) to produce the OAM wavefront and the other is the parabola phase distribution component (the second term in the right-hand side) for focusing the converted OAM wave to a point with focus length z_{focal}. The focused OAM beam generation capability of the proposed MGBM is demonstrated and exemplified in Supplementary Information F.

3.2.3. Mixed coding sequence: multi-type simultaneous reflections

To further clarify the continues scanning ability of the proposed meta-device, we seek to suitably program the MGBM to create an obliquely directed OAM vortex wavefront pointing at a pre-determined direction, as an intriguing functionality especially in massive OAM-based MIMO systems [81]. In this section, the 2-bit and 3-bit digits, "0" – "3" and "0" – "7" stand for "00" – "11" and "000" – "111", respectively. Regarding the Fourier connection between the coding patterns (in analogy to a time-domain signal) and the resultant scattering patterns (in analogy to a frequency-domain signal), diverse theorems and analytical tools in digital communication and information theory can serve for the design of coding metasurfaces, leading to various exciting functionalities. We adopt

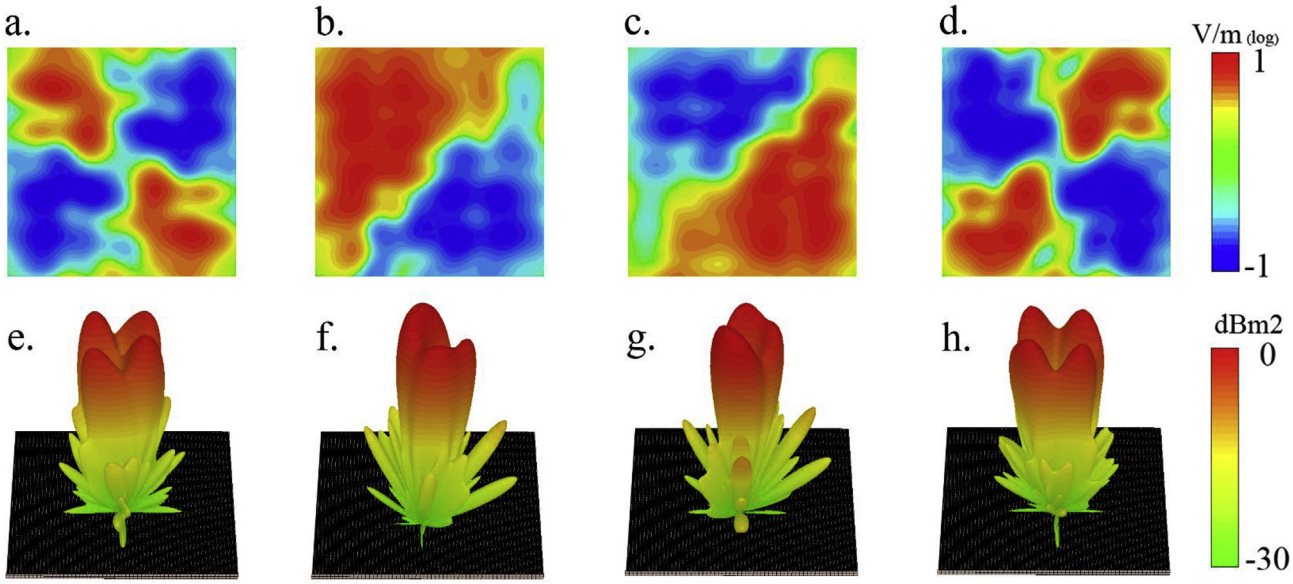

Fig. 5. (a)–(d) The phase wavefronts and (e)–(h) 3D far-field scattering patterns of the generated vortex beams with the topological charges of (a), (e) $\ell = -2$, (b), (f) $\ell = -1$, (c), (g) $\ell = +1$, and (d), (h) $\ell = +2$ under the illumination of x-polarized normal incidences at different frequencies. The numerical results are captured at $f = 1.8$ THz, $f = 1.9$ THz, $f = 2$ THz, and $f = 2.1$ THz, respectively.. (A colour version of this figure can be viewed online.)

the digital convolution theorem here to demonstrate the vortex pattern shift wherein by calculating the modulus of the spiral coding sequence and a gradient coding sequence $exp(i\lambda sin\theta_0)$, the OAM wavefront $E(sin\theta_0)$ could be arbitrarily tilted to the desired angle by a quantity of $sin\theta_0$ with little distortion [82].

$$E(x_\lambda)e^{jx_\lambda \sin\theta_0} \overset{FFT}{\longleftrightarrow} E(\sin\theta_0)*\delta(\sin\theta - \sin\theta_0) = E(\sin\theta - \sin\theta_0) \tag{11}$$

Herein, the item $\exp(i\lambda\sin\theta_0)$ describes the electric field distribution with unity amplitude and gradient phase along the desired direction. Applying the convolutional operation on MGBM follows the same concept in the Fourier transform, describing the Fourier relation between the multiplication of two signals in the time domain and the convolution of the corresponding frequency domain signals.

Such a surprising principle can be thought of as a frequency-shift function when one of the product terms as a time-shift signal turns into an impulse function in the frequency domain [82].

$$f(t)e^{j\omega_0 t} \overset{FFT}{\longleftrightarrow} f(\Omega)\delta(\Omega - \Omega_0) = f(\Omega - \Omega_0) \tag{12}$$

Equation (11) can be simply extracted by replacing arguments t and Ω in the above equation by $x_\lambda = x/\lambda$ and $sin\theta$, respectively. An intuitive sketch for demonstrating the scattering pattern shift principle through adopting the convolutional operation is schematically depicted in Fig. 6.

Importantly, the multiplication of two coding patterns, which is simply the modulus of their digital coding patterns makes a great step forward in performing more advanced missions without incurring heavy computational complexity. Several coding samples are provided here to elucidate better how the concept above can be served to elaborately modulate the direction of the OAM vortex beams. We start with a simple gradient coding sequence varying along both x and y directions whose spatial inhomogeneity deviates the impinging waves toward $\theta = 45^\circ$ and $\varphi = 150^\circ$ away from the normal direction. In this illustration, the OAM-generating spiral-type coding pattern ($\ell = -2$) is multiplied by the gradient above coding sequence, yielding the modulus of the mixed coding pattern given in Supplementary Figure S17. As can be deduced from Fig. 7a, thanks to digital convolutional property, the MGBM meta-device driven by the mixed coding pattern generates a tilted OAM scattering pattern with the topological charge of $\ell = -2$ pointing at our desired direction. The rotation angle, which is solely dictated by the special phase–gradient profile of the gradient coding sequence, is very close to our theoretical predictions based on Equations (8a) and (8b). The generality of our design can be further studied through other different prototypes with different deviation angles, topological charges, and operating frequencies, demonstrating the unprecedented degrees of freedom provided by digital codes and their addition operations. Fig. 7 displays the 2D far-field intensities of four mixed coding patterns acquired by the modulus of four different spiral-shaped digital codes and four different phase-gradient ones. Referring to 2D far-field intensities of Fig. 7 clearly proves that the original doughnut-shaped patterns carrying OAM of different modes are efficiently deflected away from the normal axis without apparent distortion, after direction. As the characteristic profile of vortex beams, the center of each emitting beam is a deep dip (about convoluted by a single-beam pattern orienting along the appointed at least 16 dB lower than the annular high-intensity region). Moreover, inconsistent with the theory [83], one can conclude that by increasing the topological charge ℓ, the diameter of ring-shaped far-field patterns also experiences a proportional increment. A slight distortion of the far-field patterns can be attributed to the non-linear relation between θ and $\sin\theta$ in Equation (11), especially for large deflection angles ($\geq 30^\circ$) [82]. Consequently, based on the concept of a scattering-pattern shift in which the coding patterns are assumed as time-domain signals, and the far-field patterns can be thought of as their corresponding frequency spectra, the versatile MGBM meta-device is capable of continues scanning by generating arbitrarily-oriented spin-controlled OAM beams. More interestingly, the versatility of our design can be further elucidated by demonstrating the MGBM's ability to carry out multiple missions, i.e., generating multi-type reflections of a differently-shaped far-field pattern at the same time.

The designed meta-device is firstly divided into $N_x \times N_y$ pixels

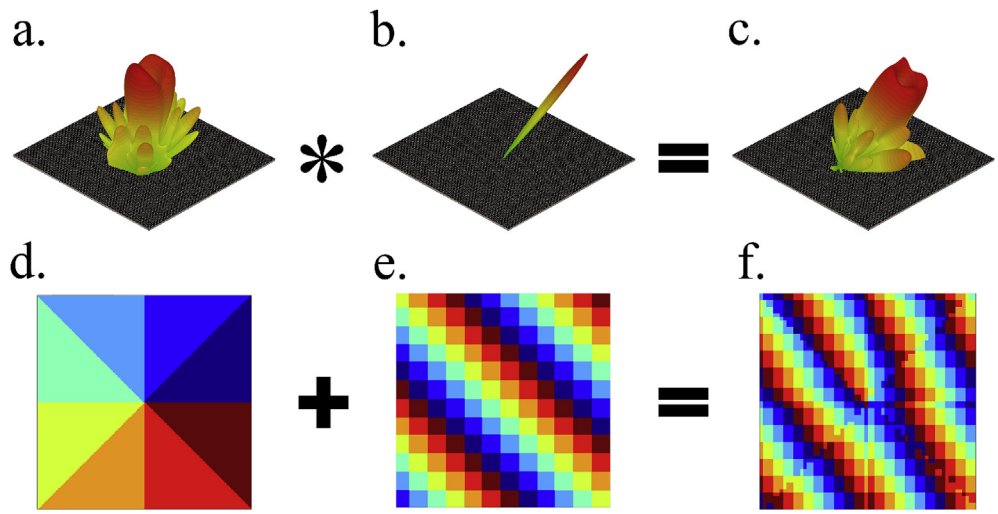

Fig. 6. Demonstration of the vortex pattern shift principle in analogy to the Fourier Transform. (a)–(c) Full-wave simulated scattering patterns corresponding to (d) the coding layout of OAM-generator metasurface, (e) the phase-gradient coding pattern, and (f) their modulus. (A colour version of this figure can be viewed online.)

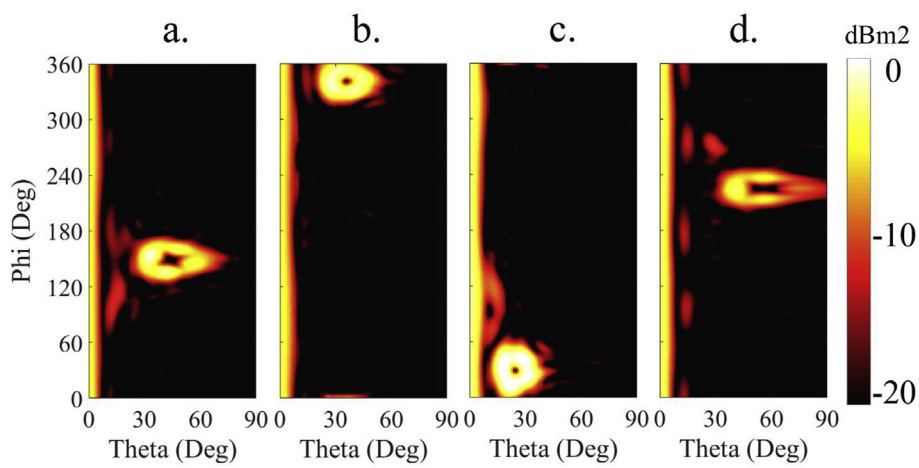

Fig. 7. The normalized 2D far-field intensities for illustration to the principle of vortex pattern shift toward different desired angles of (a) (θ_{ref}, φ_{ref})=(45°,150°), (b) (θ_{ref}, φ_{ref})=(35°,340°), (c) (θ_{ref}, φ_{ref})=(25°,30°), (d) (θ_{ref}, φ_{ref})=(55°,225°) with different topological charges of $\ell = -2$, $\ell = -1$, $\ell = +1$, and $\ell = +2$, respectively. The numerical simulations are carried out at different frequencies of $f = 2.1$ THz, $f = 1.8$ THz, $f = 2$ THz and $f = 1.9$ THz. (A colour version of this figure can be viewed online.)

each of which is occupied with one constitutive meta-atom (Fig. 1). Let us consider that the local reflection phase of the occupying meta-atoms at (x_i,y_i,0) required for producing the desired A1−type, A2−type, A3−type, …far-field patterns are denoted by $\phi_{ref}^{A_1}(x_i, y_i)$, $\phi_{ref}^{A_2}(x_i, y_i)$, $\phi_{ref}^{A_3}(x_i, y_i)$, …, respectively. To provide great convenience in diverse digital operations on the coding patterns, all digital states can be represented by unit vectors sweeping the circumference of a unit circle named as "*coding circle*". Fig. 8a-c displays the coding circle for 1-bit, 2-bit, and 3-bit digital codes, respectively. According to the coding circle, the addition theorem on the unit cell level is inspired by the parallelogram law, which gives the rule for vector addition in the well-known Euclidean geometry. For instance, a typical process of addition operation for two 2-bit digital code is demonstrated in Fig. 8d, in which 1_2+2_2 results in phase response of 135°, corresponding to the 3_3 digital state. Here, the main body is the absolute value of the digital state and the index at the bottom right corner denotes the number of bits. In accordance with the physical meaning of the holographic theory on the metasurface level, multiple independent far-field scattering patterns will appear simultaneously by incorporating the superposition of the generator digital codes (see Fig. 8e-g). In another sense, the MGBM can be re-programmed for accomplishing multiple outlandish missions at the same time that is an arduous task for prevalent metasurfaces. As can be concluded from Fig. 8h-i, the addition theorem of digital codes, which elaborately sets up a bridge between two different coding bits, was augmented by $e^{j\phi_1(m,n)} + e^{j\phi_2(m,n)} + e^{j\phi_3(m,n)} + \ldots = A(m,n)e^{j\phi_0(m,n)}$ in which φ_0(m,n) and A(m,n) are the phase and amplitude response of each meta-atom in the mixed coding pattern, respectively. Because of being normalized, the mixed coding states are described by variations of their arguments. Accordingly, the holographic theory can be served to attain the superposition of different phase profiles, enabling multi-functional property at the same time. In this case, the reflection phase of each meta-atom obeys the following relation [84].

$$\phi_{ref}(x_i, y_i) = Arg\left[\sum_i \exp\left(j\phi_{ref}^{A_i}(x_i, y_i)\right)\right] \tag{13}$$

Before we delve into the numerical simulations, we should

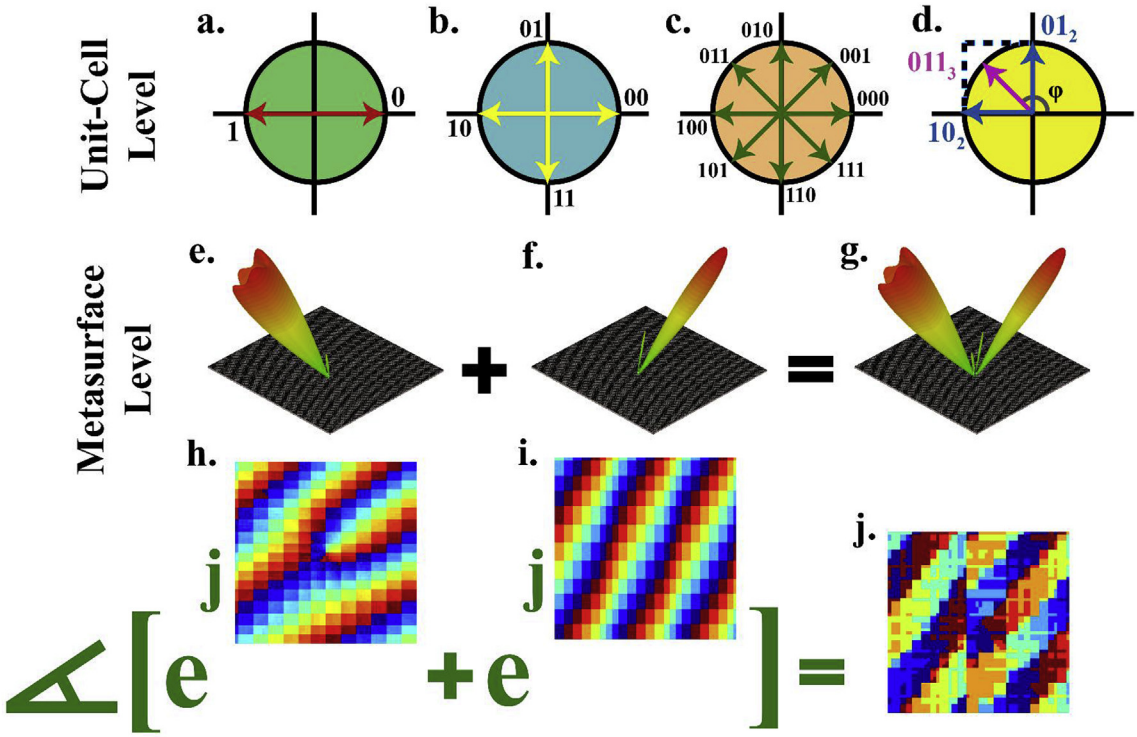

Fig. 8. Intuitive display for addition theorem on unit-cell and metasurface levels. (a)–(c) The 1-bit, 2-bit, and 3-bit complex codes and their corresponding arguments in the coding circle. (d) A typical addition process of 2-bit complex codes: $01_0+10_2=011_3$. (e) The metasurface-system level performance of addition theorem. (A colour version of this figure can be viewed online.)

notice that the above equation manifests that two different kinds of 2-bit digital codes change to 3-bit ones after involved with the addition theorem. The multifunctional MGBM can serve as a suitable platform to realize controllable multi-beam scattering patterns superimposed by using the addition theorem. Four different operational scenarios will be planned here to picture the multi-mission capability of the proposed MGBM meta-device that is dynamically interchangeable. To begin with, let us consider two 2D gradient coding sequences along different directions one of which generates single anomalously reflected beam toward ($\theta_{\text{mission1}}=20°$, $\varphi_{\text{mission1}}=15°$), while the other will divert the incident THz wavefront from the normal direction into ($\theta_{\text{mission2}}=35°$, $\varphi_{\text{mission2}}=75°$). After being modulated by the addition principle, the resultant MGBM meta-device incorporating both aforesaid digital gradient codes produces two independent scattered beams simultaneously which are successfully oriented along the predetermined directions, as illustrated in Fig. 9a. Clearly, the two beams have little interference during the addition operations. In the second example, the metasurface is designed to separate the incident plane wave into three independent pencil beams with distinct directions of ($\theta_{\text{mission1}}=30°$, $\varphi_{\text{mission1}}=40°$), ($\theta_{\text{mission2}}=30°$, $\varphi_{\text{mission2}}=120°$), and $\theta_{\text{mission3}}=30°$, $\varphi_{\text{mission3}}=230°$). Some other peculiar missions, which cannot be implemented by traditional coding metasurfaces, are also provided in Fig. 9. More interestingly, the presented concept in conjugation with the convolutional principle is utilized to generate multiple spin-controlled vortex beams of arbitrary parameters with helicoidal equal-phase wavefronts. Without loss of generality, we define a general mission in which the mixed coding pattern is calculated based on the addition theorem by incorporating the rotated OAM coding pattern ($\ell=1$) after being added with two gradient coding sequences "0 1 2 3 4 5 6 7" varying along two desired directions. Specifically, the MGBM meta-device programmed with such a mixed coding pattern (Fig. 9c) amazingly scatters the normal incidence into two asymmetric vortex beams ($\ell=1$) with the deviation angles of ($\theta_{\text{mission1}}=40°$, $\varphi_{\text{mission1}}=135°$) and ($\theta_{\text{mission2}}=25°$, $\varphi_{\text{mission2}}=330°$), respectively. The 3D far-field scattering patterns displayed in Fig. 9c bring out the excellent effect that two differently-oriented vortex beams flourishingly occur at the same time. In the other mixed coding sample, when two phase-gradient and spiral coding sequences are superimposed together by using the complex addition operation, we will obtain a tilted OAM beam with ℓ=1 and a single pencil beam pointing at ($\theta_{\text{mission1}}=25°$, $\varphi_{\text{mission1}}=250°$) and ($\theta_{\text{mission2}}=30°$, $\varphi_{\text{mission2}}=25°$), respectively. Theoretically speaking, the deviation angles of all scattering beams are in an excellent match to our predictions. The above numerical results imply that the proposed MGBM meta-device can efficaciously carry out the multiple missions were assigned to it. In overall, all multi-type scattering phenomena are permanently available in a real-time manner within one metasurface. The presented multi-functional performances including pencil beam–pencil beam, vortex beam–vortex beam, and vortex beam–pencil beam far-field intensities are of paramount importance especially in multi-channel information modulation as well as channel multiplexing in the communication systems. The multiple vortex beams carrying OAMs with different modes can be used to encode information and increase the communication capacity without increasing the bandwidth.

Once again, we should remember that the MGBM meta-device could be instantaneously switched between all exciting functionalities discussed throughout this paper by simply changing the spatial distribution of biasing voltages provided by a computer-controlled voltage source. This feature is guaranteed by using a biasing grid providing the required DC voltages for each meta-atom, separately. The schematic representation of the biasing lines can be found in Supplementary Figure S18. Meanwhile, the

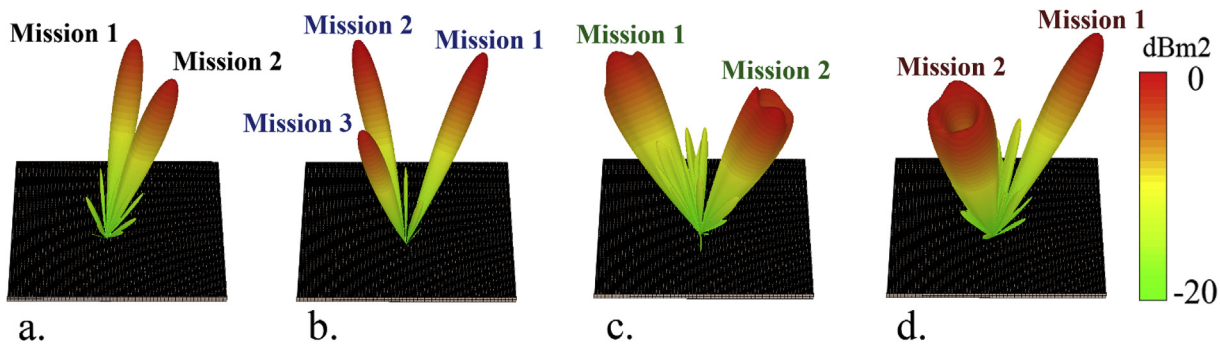

Fig. 9. The 3D far-field scattering patterns of the mixed coding layouts to illustrate the concept of multi-functionality of the MGBM meta-device arising from the addition at $f = 2$ THz. It is shown that under the illumination of a y-polarized normal incidence, the MGBM structure is capable of accomplishing different missions in which the normal plane wave is reflected into (a) two distinct pencil beams pointing at ($\theta_{\text{mission1}} = 20°$, $\varphi_{\text{mission1}} = 15°$) and ($\theta_{\text{mission2}} = 35°$, $\varphi_{\text{mission2}} = 75°$); (b) three distinct pencil beams pointing at ($\theta_{\text{mission1}} = 30°$, $\varphi_{\text{mission1}} = 40°$), ($\theta_{\text{mission2}} = 30°$, $\varphi_{\text{mission2}} = 120°$), and ($\theta_{\text{mission3}} = 30°$, $\varphi_{\text{mission3}} = 230°$); (c) two distinct vortex beams with the topological charge of ℓ=1 pointing at ($\theta_{\text{mission1}} = 40°$, $\varphi_{\text{mission1}} = 135°$) and ($\theta_{\text{mission2}} = 25°$, $\varphi_{\text{mission2}} = 330°$); (d) two distinct pencil- and vortex-type (ℓ=1) beams pointing at ($\theta_{\text{mission1}} = 30°$, $\varphi_{\text{mission1}} = 25°$) and ($\theta_{\text{mission1}} = 25°$, $\varphi_{\text{mission1}} = 250°$), respectively. (A colour version of this figure can be viewed online.)

effects of these lines on the performance of the individual meta-atoms and the beam-deflecting and OAM-generating MGBMs are discussed in Supplementary Information H. The results show that the MGBMs still work correctly in the presence of the employed golden biasing lines. Due to the recent development of large-scale graphene synthesis, transfer techniques, and its compatibility with the existing integrated electronics platform, the feasibility of our design can be further guaranteed [85]. Compared with previous designs, the proposed MGBM allows an unprecedented degree of freedom in wave manipulation and is much easier to be integrated into the THz devices, implying significant scientific value in practical applications.

4. Conclusion

In summary, beyond the scope of fixed-function coding metasurfaces, for the first time, we have designed a re-programmable MGBM at THz frequencies that can perform different missions from controllable OAM wavefront generation to continuous scan of the single-beam radiation in a real-time manner. The multi-functionality of the proposed MGBM meta-device was guaranteed by designing graphene-based digital meta-atoms whose operational statuses can be instantaneously switched among eight states of "000", "001", "010", "011", "100", "101", "110", and "111" by mere changing the voltage configurations imparted on the graphene patterns. The versatility of the MGBM structure was further demonstrated by employing different arrangements of digital meta-atom scheduling phase-gradient, spiral-like, and spiral-parabola-like coding schemes. It was also shown that different from earlier works, by applying a multi-voltage external source to the gated structure controlled by an FPGA platform, the designed MGBM has the ability to dynamically reconfigure the specifications and directions of emitting angular momentum and anomalously reflected beams. The simulation results illustrate very good agreement with our theoretical predictions, where the addition theorem and convolutional principle in analogy to Fourier Transform were exploited to indicate the excellent performance of the proposed MGBM for realizing multi-type reflections at the same time, such as multiple pencil beams, multiple vortex beams with different OAM modes, and multiple pencil–vortex beams. If the other wave-manipulation functionalities are desired, one may easily replace Equations (8a) and (8b) by the phase distributions pertaining to those functionalities. The simple and integrated structural design of the proposed MGBM may be a key in enabling future intelligent THz-wave devices with reconfigurable missions that open a pathway to novel types of multi-beam generation and provide a wide platform for applications involving imaging, tunable display, camouflage technology, and information systems.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.carbon.2019.04.034.

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